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CSE 3025 Lab 3 Report

Part 1:

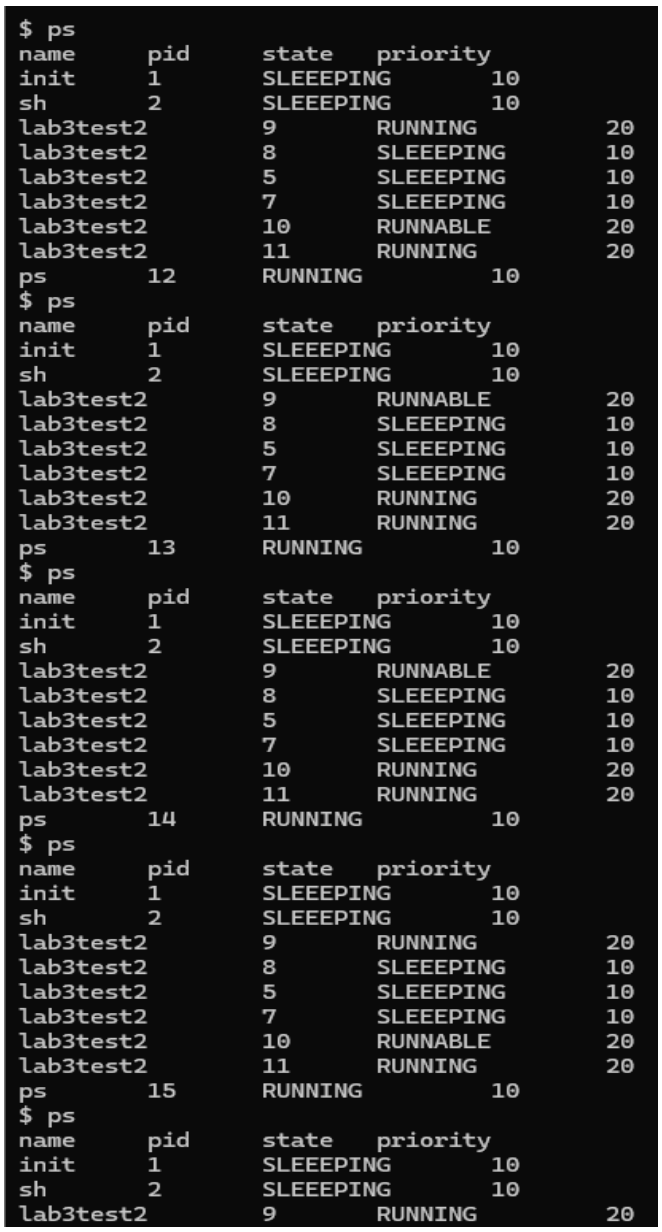
```
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
ps       3       RUNNING         10
$
```

Part 2:

```
$ lab3test 4 &
$ Parent C2hild1  c22 crreaeatiteng d
child 22
ParenCt 21hil cdre a23ti cng reactehidld
23
Parent 21 creating child 24
Parent 21 creating child 25
Child 24 created
ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test         22      RUNNING         20
ps       26      RUNNING         10
lab3test         23      RUNNABLE        20
lab3test         24      RUNNING         20
lab3test         25      RUNNABLE        20
$ nice 5 25
Unrecognized process...
$ nice 25 5
Child 25 created
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test         22      RUNNING         20
ps       29      RUNNING         10
lab3test         23      RUNNABLE        20
lab3test         24      RUNNABLE        20
lab3test         25      RUNNING         5
```

Part 3:

As can be seen with the screenshot below, the processes do seem to be rotating mainly between "runnable" to "running", indicative of the Round Robin scheduling working.



```
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test2        9       RUNNING         20
lab3test2        8       SLEEEPING       10
lab3test2        5       SLEEEPING       10
lab3test2        7       SLEEEPING       10
lab3test2        10      RUNNABLE        20
lab3test2        11      RUNNING         20
ps       12      RUNNING         10
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test2        9       RUNNABLE        20
lab3test2        8       SLEEEPING       10
lab3test2        5       SLEEEPING       10
lab3test2        7       SLEEEPING       10
lab3test2        10      RUNNING         20
lab3test2        11      RUNNING         20
ps       13      RUNNING         10
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test2        9       RUNNABLE        20
lab3test2        8       SLEEEPING       10
lab3test2        5       SLEEEPING       10
lab3test2        7       SLEEEPING       10
lab3test2        10      RUNNING         20
lab3test2        11      RUNNING         20
ps       14      RUNNING         10
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test2        9       RUNNING         20
lab3test2        8       SLEEEPING       10
lab3test2        5       SLEEEPING       10
lab3test2        7       SLEEEPING       10
lab3test2        10      RUNNABLE        20
lab3test2        11      RUNNING         20
ps       15      RUNNING         10
$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
lab3test2        9       RUNNING         20
```

Part 4:

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ lab3test2 &; lab3test2 &; lab3test2 &
$ ParCenht 5 crild ea9 createPd
arent 8 creatingP archent il7 dcre 10ati
ntg inchg ilchd 1il1d 9

CChihild ld 1110 cr ceatreedate
d

$ ps
Child 10 terminated
Child 11 terminated
name     pid     state   priority
init     1       RUNNING         10
sh       2       SLEEEPING       10
lab3test2        9       RUNNING         20
lab3test2        5       SLEEEPING       10
ps       12      RUNNING         10
C$ hild 9 terminated

$ ps
name     pid     state   priority
init     1       SLEEEPING       10
sh       2       SLEEEPING       10
ps       13      RUNNING         10
```

This output shows that priority scheduling works because the ps command had a priority of 10 and so did the parent processes of the lab3test. So when we ran the test and then ran ps it waited until the parent processes were finished utilizing a first come first serve scheduling due to the processes having the same priority before outputting the current processes and their statuses. It then showed the child processes running which again aligns with the priority scheduling we implemented because the child processes had a priority of 20 which meant that they would only run after ps and the parent processes would be finished. This is then highlighted by the output we see above.